

Supplementary Materials for **Nonlinear climate sensitivity and its implications for future greenhouse warming**

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Supplementary Materials

table S1. Long-term (784 to 10 ka B.P.) paleorecords of SST. Names, locations, SST-reconstruction methods and references for the 14 long-term SST-records covering the period 784 – 10 ka B.P.

Core Name	Location	SST proxy	Reference
North Atlantic:			
ODP982	57.31°N / 16.52°W	alkenone	(53)
IODP306-U1313	41.00°N / 32.96°W	alkenone	(54)
South Atlantic:			
ODP175-1082	21.09°S / 11.82°E	alkenone	(55)
Indian Ocean:			
ODP722	16.62°N / 59.80°E	alkenone	(56)
South China Sea:			
ODP184-1143	9.36°N / 113.29°E	alkenone	(57)
IODP1146	19.46°N / 116.27°E	alkenone	(56)
Western Pacific			
MD97-2140	2.03°N / 141.77° E	Mg/Ca	(58)
ODP806b	0.32°N / 159.36°E	Mg/Ca	(59)
MD06-3018	23.00°S / 166.13°E	Mg/Ca	(60)
Central Pacific:			
RC11-210	1.82°N / 140.10°W	radiolarian faunal analysis	(61)
Eastern Pacific			
ODP846	3.10°S / 90.82°W	alkenone	(62)
Southern Ocean:			
DSDP90-594	45.52°S / 174.95°E	MAT	(63)
ODP181-1123	41.79°S / 171.50°E	MAT	(64)
ODP177-1090	42.92°S / 8.90°E	alkenone	(65)

table S2. All paleorecords of SST used in this study. Names, locations, SST-reconstruction methods and references for the all SST-records used in the study.

Core Name	Location	SST proxy	Reference
North Atlantic:			
ODP982	57.31°N / 16.52°W	alkenone	(53)
IODP306-U1313	41.00°N / 32.96°W	alkenone	(54)
SU92-03	43.20°N / 10.11°W	MAT	(66)
MD95-2042	37.80°N / 10.17°W	MAT	(66)
ODP165-999A	12.74°N / 78.74°W	MAT	(67)
MD95-2040	40.58°N / 9.86°W	MAT	(68)
MD02-2575	29.00°N / 87.12°W	Mg/Ca	(69)
ODP165-1002	10.71°N / 65.17°W	alkenone	(70)
MD95-2037	37.09°N / 32.03°W	alkenone	(71)
MD01-2444	37.57°N / 10.13°W	alkenone	(72)
MD03-2707	2.50°N / 9.39°E	Mg/Ca	(73)
K708-1	50.00°N / 23.73°W	MAT	(74)
Mediterranean Sea:			
ODP161-977A	36.03°N / 1.96°W	alkenone	(75)
South Atlantic:			
ODP175-1082	21.09°S / 11.82°E	alkenone	(55)
GeoB1028-5	20.10°S / 9.19°E	alkenone	(76)
GeoB1016-3	11.77°S / 11.68°E	alkenone	(76)
GeoB1008-3	6.58°S / 10.32°E	alkenone	(76)
GeoB1712-4	23.26°S / 12.80°E	alkenone	(77)
GeoB1710-3	23.43°S / 11.70°E	alkenone	(77)
GeoB1112	5.77°S / 10.75°W	Mg/Ca	(78)
GeoB1105	1.67°S / 12.43°W	Mg/Ca	(78)
Indian Ocean:			

ODP722	16.62°N / 59.80°E	alkenone	(56)
GeoB3007-1	16.17°N / 59.76°E	alkenone	(79)
T93929-P	13.70°N / 53.25°E	alkenone	(80)
MD90-0963	5.07°N / 73.88°E	alkenone	(80)
MD85674	3.18°N / 50.43°E	alkenone	(80)
MD85668	0.02°S / 46.04°E	alkenone	(80)
South China Sea:			
ODP184-1143	9.36°N / 113.29°E	alkenone	(57)
IODP1146	19.46°N / 116.27°E	alkenone	(56)
GIK17954-2	14.77°N / 111.53°E	alkenone	(81)
GIK17961-2	8.51°N / 112.33°E	alkenone	(81)
ODP1144	20.05°N / 117.42°E	Mg/Ca	(82)
ODP1145	19.58°N / 117.63°E	Mg/Ca	(83)
Western Pacific			
MD97-2140	2.03°N / 141.77° E	Mg/Ca	(58)
ODP806b	0.32°N / 159.36°E	Mg/Ca	(59)
MD06-3018	23.00°S / 166.13°E	Mg/Ca	(60)
MR97-04-1MUC	40.56°N / 142.93°E	MAT	(84)
MD01-2421	36.02°N / 141.78°E	MAT	(84)
MR00-05-2PC	40.00°N / 146.00°E	MAT	(84)
MR99-04-3PC	37.50°N / 152.00°E	MAT	(84)
MR02-03-2PC	36.00°N / 146.50°E	MAT	(84)
ODP871	5.56°N / 172.34°E	Mg/Ca	(85)
MD05-2920	2.51°S / 144.32°E	Mg/Ca	(14)
Central Pacific:			
RC11-210	1.82°N / 140.10°W	radiolarian faunal analysis	(61)
W8402A-14	0.95°N / 138.96°W	alkenone	(86)

Eastern Pacific			
ODP846	3.10°S / 90.82°W	alkenone	(62)
ODP202-1240	0.02°S / 86.46°W	Transfer function	(87)
GeoB3388-1	25.22°S / 75.53°W	alkenone	(88)
MD02-2529	8.21°N / 84.12°W	alkenone	(89)
ODP1012	32.28°N / 118.38°W	alkenone	(90)
TR163-19	2.26°N / 90.95°W	Mg/Ca	(91)
ODP1014	32.80°N / 118.90°W	alkenone	(92)
Southern Ocean:			
DSDP90-594	45.52°S / 174.95°E	MAT	(63)
ODP181-1123	41.79°S / 171.50°E	MAT	(64)
ODP177-1089	40.94°S / 9.89°E	MAT	(93)
ODP177-1090	42.92°S / 8.90°E	alkenone	(65)
GeoB3327-5	43.24°S / 79.99°W	alkenone	(88)
PS2489-2	42.87°S / 8.79°E	MAT	(94)
MD88-770	46.02°S / 96.46°E	transfer function	(95)
MD84-551	55.01°S / 73.17°E	transfer function	(96)
U939	44.53°S / 179.50°E	alkenone	(97)
MD97-2120	45.53°S / 174.93°E	alkenone	(98)
PS75/034-2	54.37°S / 80.09°W	alkenone	(88)

table S3. Warm-phase specific equilibrium climate sensitivities and corresponding global mean surface temperature change at year 2100. Impact of different estimates of preindustrial-Last Glacial Maximum (PI-LGM) global mean SAT change on specific equilibrium climate sensitivity and future greenhouse warming. Warm-phase specific equilibrium climate sensitivity (S^{warm}), warm-phase transient climate response parameter (TCRP) and resulting year 2100 global mean surface air temperature (SAT) change (with respect to preindustrial value) based on an estimate of ocean heat uptake efficiency and the RCP8.5 radiative forcing timeseries.

	Averaged, reconstructed global mean SAT-anomaly	Global mean SAT-anomaly reconstructed from SST-proxies	Global mean SAT-anomaly reconstructed from model simulation	Averaged, reconstructed global mean SAT-anomaly scaled to 3.0 K PI-LGM difference (3)	Averaged, reconstructed global mean SAT-anomaly scaled to 4.0 K PI-LGM difference (21)	Global mean SAT-anomaly reconstructed from Antarctic temperature anomaly (30) using a polar amplification factor of 1.2 (31)	Global mean SAT-anomaly reconstructed from Antarctic temperature anomaly (30) using a polar amplification factor of 1.9 (31)
PI-LGM global mean SAT anomaly (K)	5.75	5.00	6.50	3.00	4.00	7.71	4.68
S^{warm} ($\text{K W}^{-1} \text{m}^2$)	1.32	1.40	1.24	0.68	0.91	1.52	0.97
Warm-phase TCRP ($\text{K W}^{-1} \text{m}^2$)	0.74	0.76	0.71	0.48	0.59	0.80	0.61
Year 2100 global mean SAT change (K)	5.86	6.00	5.65	3.84	4.68	6.32	4.87

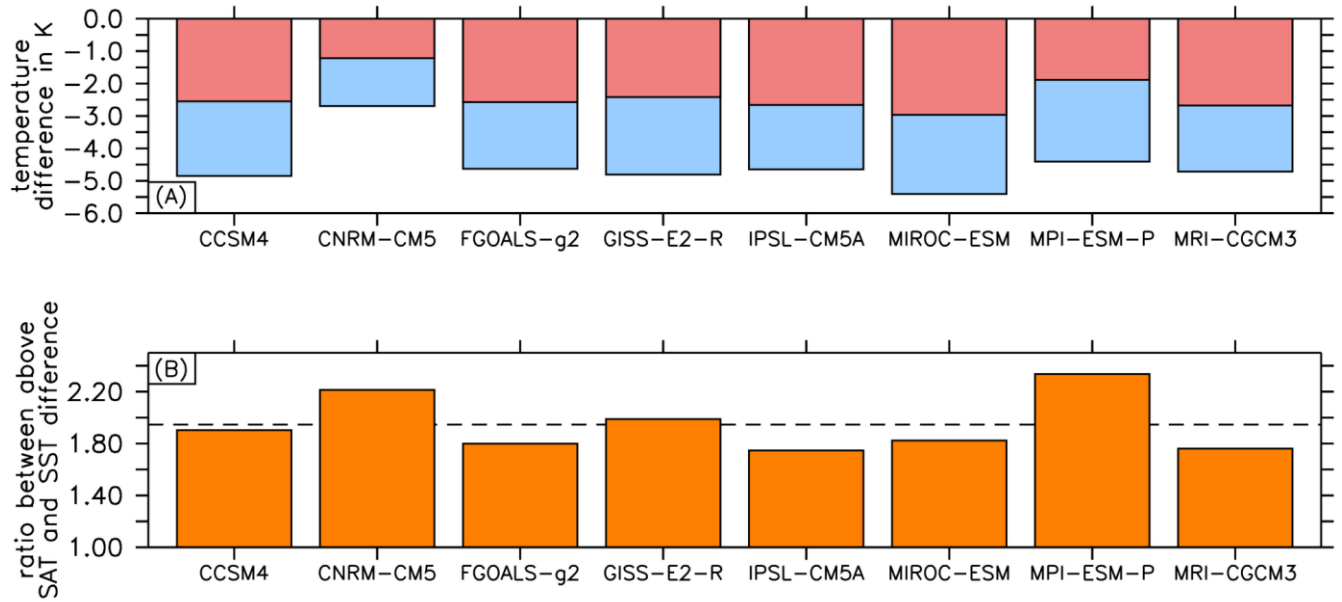


fig. S1. Underestimation factor for SST-proxy network. (A): Simulated LGM-PI global mean SAT-difference for 8 PMIP3 models (K, blue bars) (13). Averaged simulated LGM-PI SST-difference (K) for SST subsampled at long-term paleo-record locations in the respective PMIP3 model (pink bars). **(B):** Ratio of LGM-PI differences shown in (A). Dashed line indicates the average of all bars.

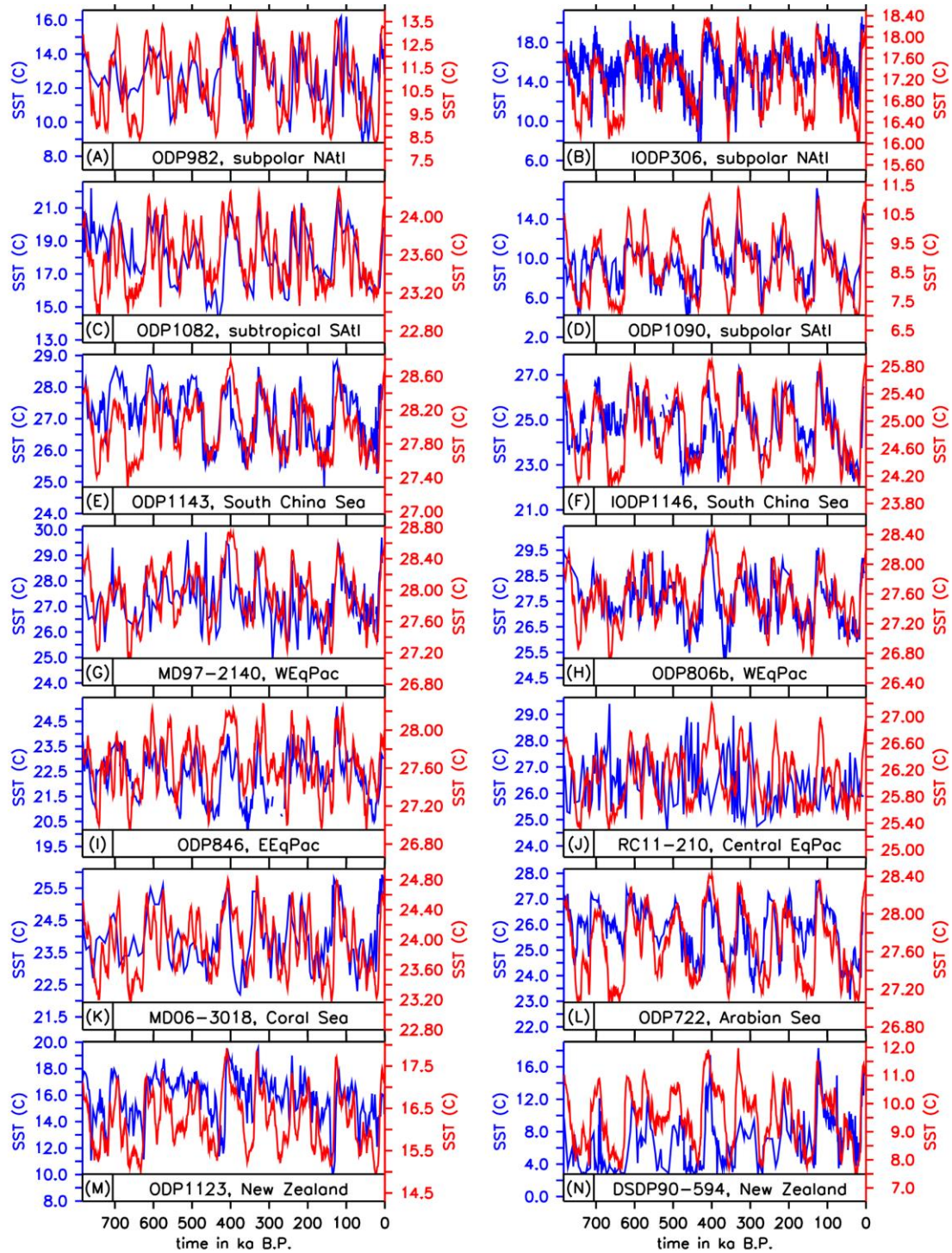


fig. S2. Model-data comparison of SST-anomalies for 14 SST proxy locations. Blue lines and left-hand axis: reconstructed SST in °C; red lines and right-hand axis: simulated SST in °C. Cores as indicated in the panels. Please refer to Table S1 for details regarding the cores and to Fig. 1 of the main manuscript for core locations. For simulated SST, values of the grid cell closest to the core location were taken and 1000-year averages were calculated.

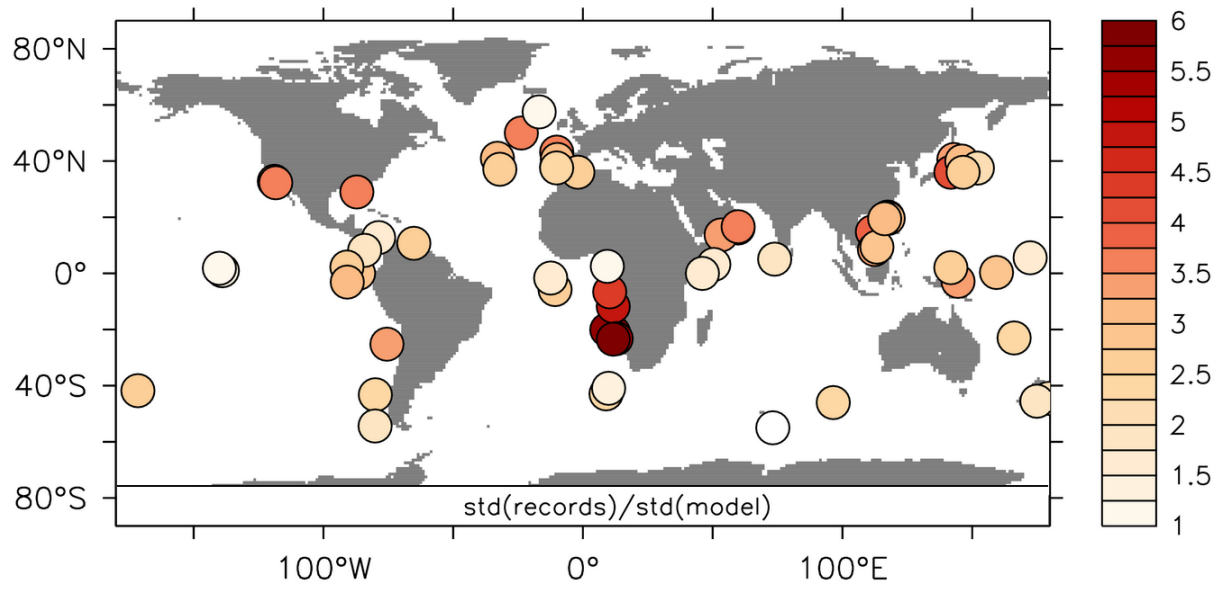


fig. S3. Ratios of standard deviations of proxy-based and simulated SST-anomalies. Simulated SST was subsampled at the 63 core locations provided in table S2 and 1000-year averages were calculated.

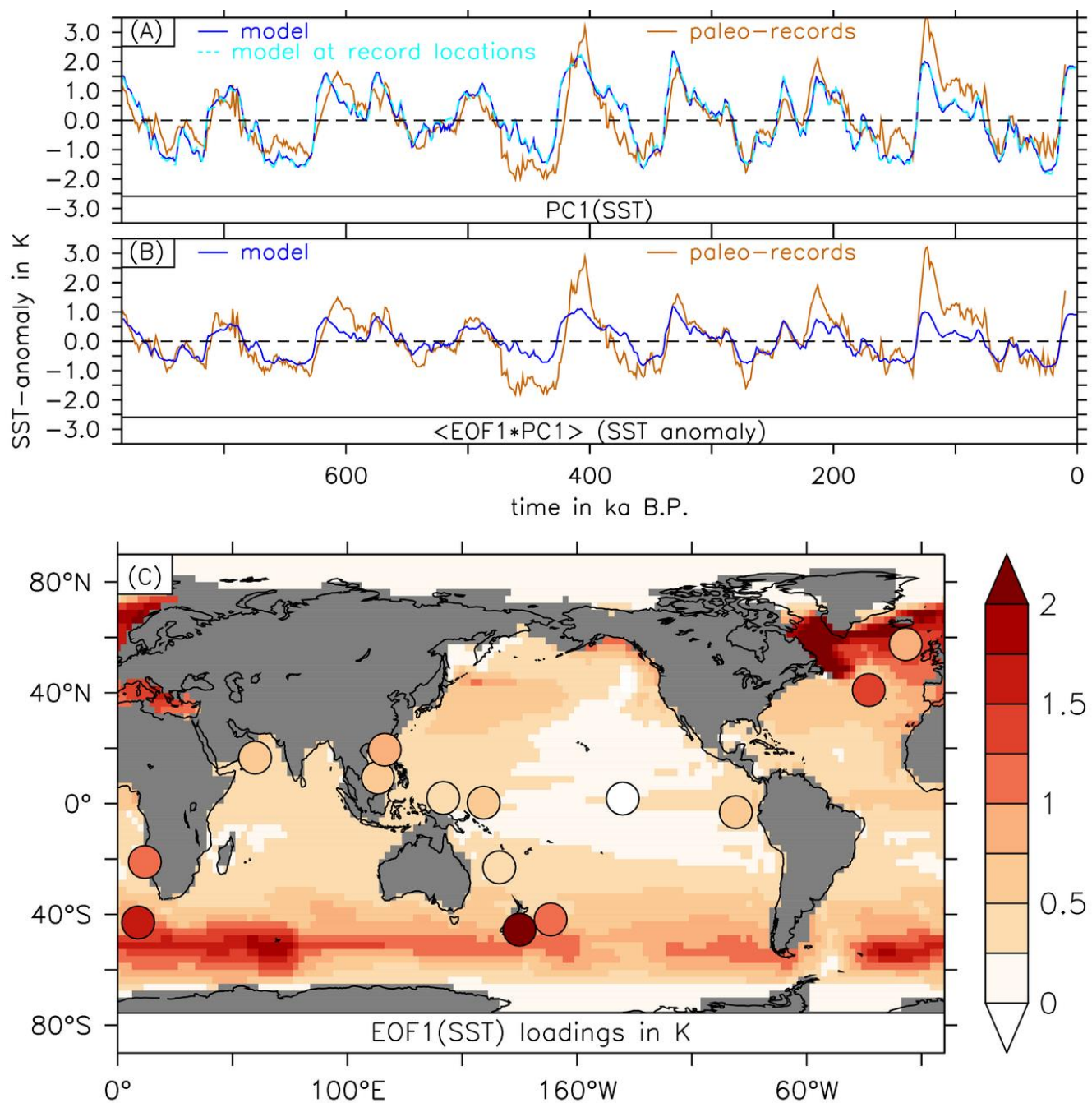


fig. S4. Pattern and temporal evolution of leading EOF1 of reconstructed and simulated SST. (A): Principal components (PC) of the EOF1 for SST from 14 long-term paleo-records (orange) and simulated SST (blue) using every model grid point and using only model grid points that correspond to the locations of the 14 long-term paleo-records (cyan dashed). **(B):** Globally-averaged SST anomaly from the EOF1 based reconstruction. Colors as in (A). **(C):** EOF1-pattern (K) for 14 long-term paleo-records (circles) and simulated SST in global domain (shading).

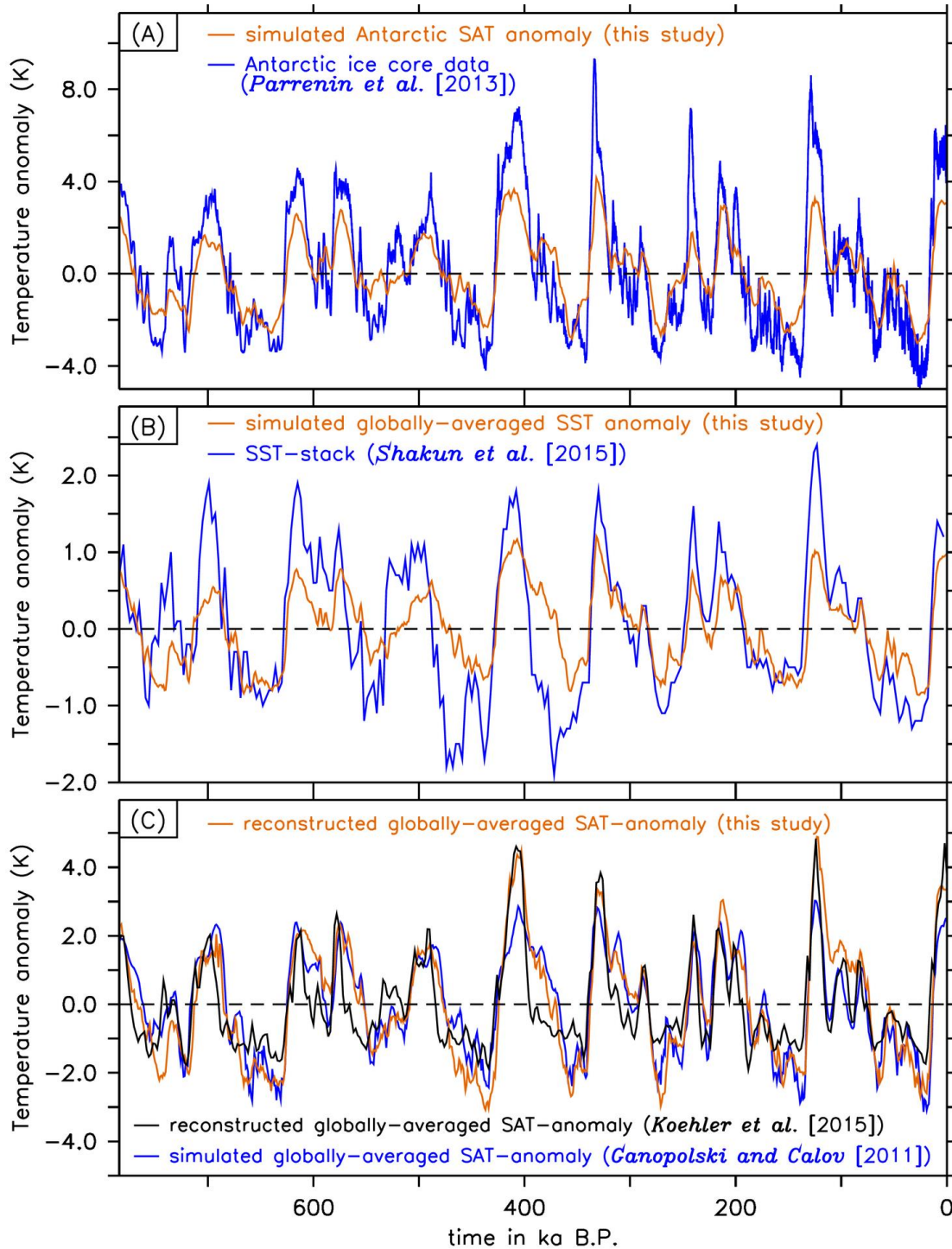


fig. S5. Comparison of long-term temperature estimates. Temperature anomalies (K) were calculated with respect to their individual averages. **(A):** Data composite of Antarctic temperature (30) (blue) and simulated Antarctic Surface Air Temperature (orange). **(B):** SST-stack (41) (blue) and simulated, globally-averaged SST (orange). **(C):** Reconstructed, globally-averaged SAT-anomaly from ref(7) (black) and this study (orange) and simulated globally-averaged SAT-anomaly (35).

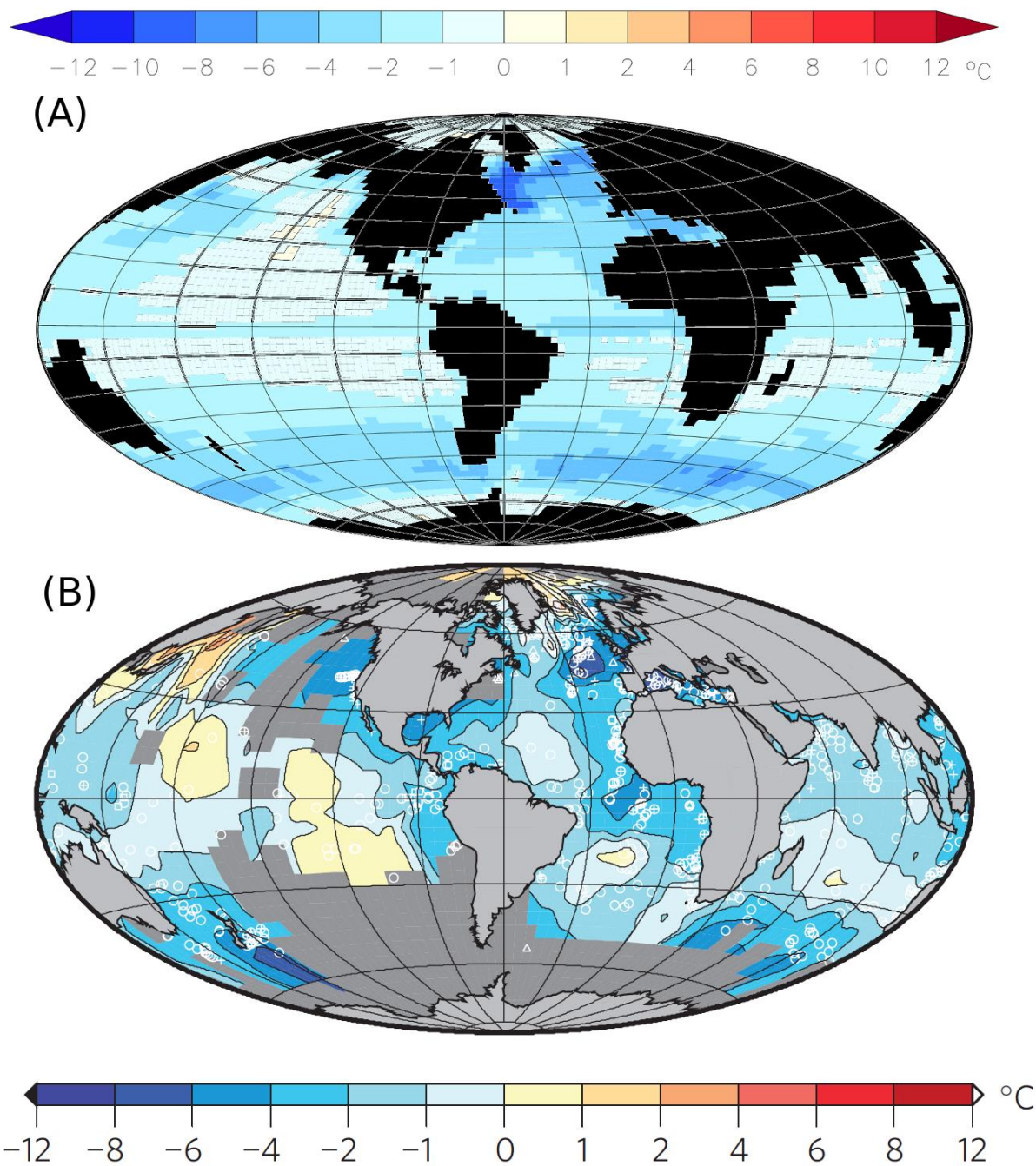


fig. S6. Model-data comparison of glacial-interglacial SST anomalies. (A): Simulated SST anomaly in K. **(B):** SST anomaly in K from the Multiproxy Approach for the Reconstruction of the Glacial Ocean (MARGO) (40).

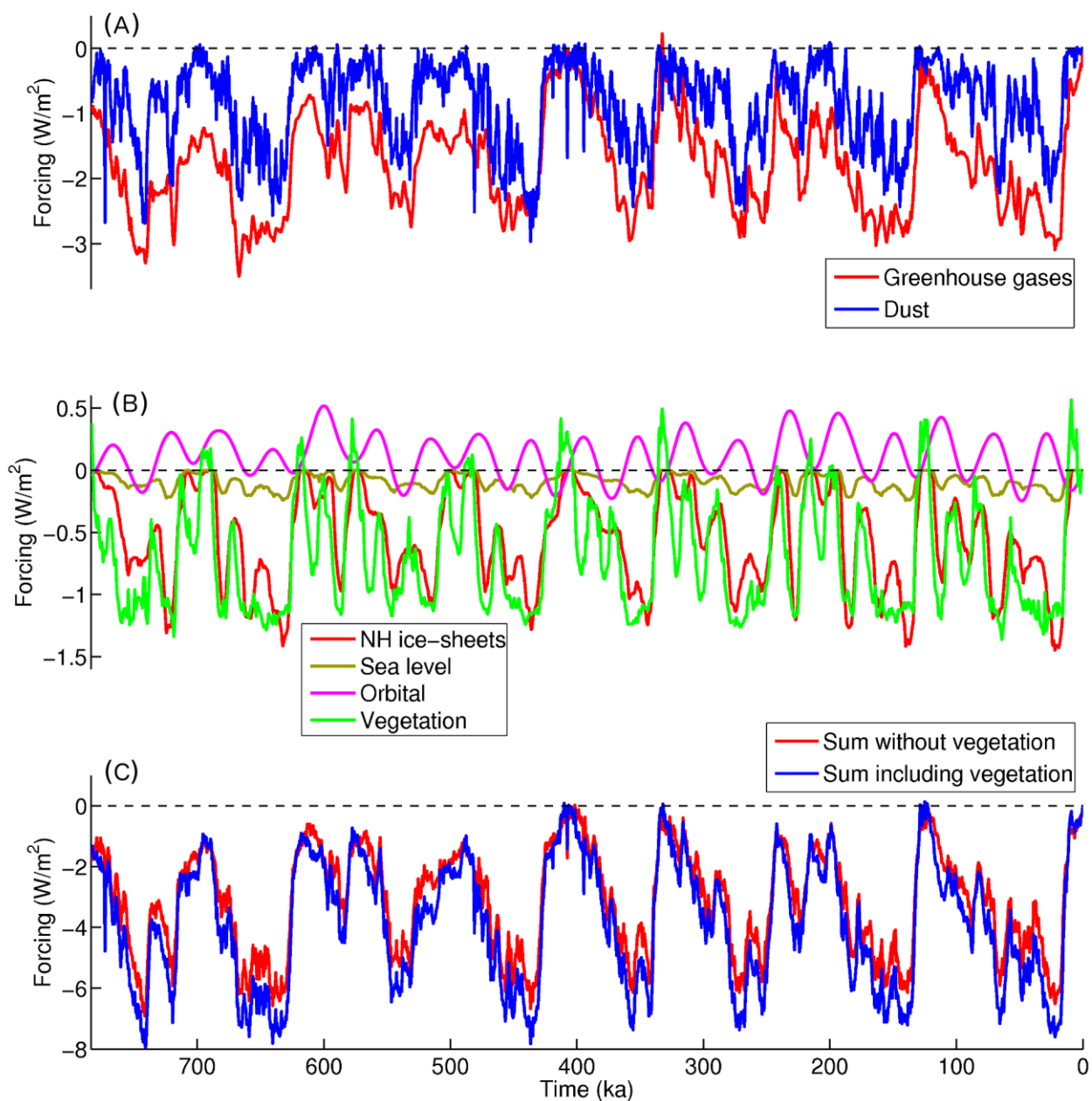


fig. S7. Radiative forcing estimates (W/m^2). (A): Greenhouse gas and dust forcings, calculated from ref.(2) and ref.(1) respectively. (B): Shortwave forcings due to changes in ice-sheet albedo, continental shelf albedo (sea level effect), orbital forcing and vegetation. (C): Sum of all forcings with and without vegetation.